

Bradon Thymes

☎ 610-241-0651 | ✉ bmt.wcpa@gmail.com | 🌐 [bradonthymes](https://bradonthymes.com) 📱 [Bradon Thymes](https://bradonthymes.com)

EDUCATION

Cornell University

Ph.D. Candidate in Computer Science

Expected graduation: May 2028

Ithaca, NY

Howard University, GPA: 3.74 (Magna Cum Laude)

Bachelor of Science in Computer Science

Aug. 2018 – May 2022

Washington, DC

PUBLICATIONS & PREPRINTS

Peer-Reviewed / Published

1. S. Shaar*, **B. Thymes***, S. Chaixanien, C. Cardie, B. Hariharan. “MovieRecapsQA: A Multimodal Open-Ended Video Question-Answering Benchmark.” **CVPR** 2026. [[arXiv](#)]
2. W. Guo, A. Wang, **B. Thymes**, T. Joachims. “Ranking with Slot Constraints.” **KDD** 2024, pp. 956–967. [[DOI](#)]
3. J. Lee, **B. Thymes**, J. Zhou, T. Joachims, R. Kizilcec. “Augmenting Holistic Review in University Admission using Natural Language Processing for Essays and Recommendation Letters.” 2023. [[arXiv](#)]
4. N. Scott, **B. Thymes**, J. Salisbury. “Statistical Modelling of Air-Ground Remotely Sensed Geo-Intelligence Information Using Naïve Bayesian Classification: A Decision-Making Approach.” 2023. [[DOI](#)]
5. N. Scott, **B. Thymes**, J. Salisbury. “Dempster–Shafer Evidential Theory Belief Amalgamation and Dynamic Programming Supporting Soldier Squadron Adversarial Engagement: Simulation-Based Decision-Making.” 2023. [[DOI](#)]
6. M. Munje, L. Teran, **B. Thymes**, J. Salisbury. “TEAM3 Challenge: Tasks for Multi-Human and Multi-Robot Collaboration with Voice and Gestures.” [[ACM](#)]
7. L. Van Dammen, N. Barnett, R. Conrady, L. Wright, **B. Thymes**, E. Shirtcliff. “Evoking Stress Reactivity in a Virtual Dance Competition.” *Advances in Simulation and Digital Human Modeling*, vol. 1206, 2020, pp. 48–55. [[PDF](#)]

Preprints

1. S. Dooley, R. Downing, G. Wei, N. Shankar, **B. Thymes**, et al. “Comparing Human and Machine Bias in Face Recognition.” [[arXiv](#)]

* *Equal contribution.*

RESEARCH EXPERIENCE

Cornell University

Graduate Research Assistant

Aug. 2022 – Present

Ithaca, NY

- Putting a Face to the Name: Named Character Grounding for Long-Form Video Understanding
 - * Built a pipeline to ground character names to their face tracks in long-form video using WhisperX sentence-level timestamps, LLM-based introduction/referent event extraction, and active speaker detection (ASD).
 - * Designed an LLM-based extraction system to automatically identify self, second-person, and third-person referrals from WhisperX-generated transcripts, reducing reliance on scripts/cast lists.
 - * Implemented local visual grounding by extracting 10 second clips around name mentions, overlaying ASD annotations and using visual prompting with an MLLM to infer referents.
 - * Building an identity-aware VideoQA benchmark as a downstream evaluation; generating questions that require “who-is-who” reasoning (eg. speaker identity, addressee resolution, etc.) and measuring how explicit character grounding improves VideoQA and long-form understanding
- MovieRecapsQA: A Multimodal Open-Ended Video Question-Answering Benchmark

- * Co-created MovieRecapsQA, an open-ended VideoQA benchmark build from movie recap videos and summaries, aligned with the full-length movie and subtitles, spanning 60 films with 8231 QA pairs and 16462 aligned atomic facts for verification.
- * Introduced a reference-free, fact-grounded evaluation for open-ended VideoQA: derive atomic facts per question and use an LLM judge to score the factuality, relevance, and coherence of the model answer.
- * Benchmarked 7 state-of-the-art models on our dataset (proprietary and open-source) plus a human study; found vision-centric questions are hardest, models over-rely on dialogue, and dialogue-only often outperforms multimodal questions.
- Multilabel Classification of Student Admission Essays
 - * Designed and implemented a multilabel classification system to extract attributes from student admission essays, utilizing T5 for generating tags.
 - * Troubleshoot and addressed challenges with imbalanced labeling across 30 attributes, where only 5 labels were significantly represented.
 - * Analyzed and optimized preprocessing techniques to improve label representation and mitigate imbalanced data issues.
 - * Gained valuable experience in handling imbalanced datasets and understanding the practical challenges of applying NLP to real-world problems.

Riverside Research

May 2022 – Aug. 2022

Machine Learning Intern (GEM Fellow)

Lexington, MA

Worked on research project involving human-robot collaboration using the game TEAM3.

- * Created a dataset for the TEAM3 blocks placed in different orientations
- * Finetuned a pretrained ResNet50 model to detect TEAM3 blocks in different orientations
- * Contributed to a set of path planning algorithms for Sawyer robotic arm to build block towers

Amazon Voice FAQ

Sep. 2021 – May 2022

Research Intern

Washington, DC

Joint research project with Howard University and Amazon which sought to assess cognitive decline through vocal prosody and facial expressions

- * Utilized PyAudio to record audio from the users
- * Reduced noise in audio by utilizing SciPy to create a high-pass filter
- * Utilizing the SpeechBrain toolkit to detect intonation of the users

University of Maryland Combinatorics & Algorithms for Real Problems

June 2021 – May 2022

Research Intern (REU-CAAR)

Remote

Worked on two separate research projects over the course of the summer:

Bias in Facial Recognition:

- * Conducted research to gain a better understanding of what biases facial recognition models have compared to human biases
- * Participated in curating a dataset from the Labeled Faces in the Wild and Celeb A facial recognition datasets
- * Labelled each individual's Fitzpatrick skin type, country of origin, birth date, as well as, their gender presentation
- * Worked on a Flask application to test human user's accuracy and biases compared to facial recognition systems
- * Worked on user verification test, where a person would compare two images to one another and determine if that person was the same individual
- * Worked on the user details section, where the participant identified their age, gender presentation, and Fitzpatrick skin type
- * Worked on website organization, using Flask sessions to route user to the verification, identification and user-details pages

Organizational Team Formation:

- * Worked on formulating a problem statement
- * Reviewed psychology team formation literature to determine the best way to optimize teams in a real-world environment

Iowa State University SPIRE-EIT

Virtual Reality Research Intern

May 2019 – Aug. 2019

Ames, IA

Through a Research Experiences for Undergraduates (REU) program, I worked with the Stress Physiology Investigative Team (SPIT) Lab at Iowa State University. Our task was to create a stress response in virtual reality:

- * Oversaw the experiment on all eighteen participants, including confirming data accuracy by running a control task
- * Researched psychology publications to identify and determine causes of increased cortisol levels
- * Programmed the music manager, light manager and audience in C# to create a task where the user had to perform a dance in virtual reality meant to trigger a stress response in virtual reality
- * Implemented dancing animations and scripts using Unity animator controllers
- * Analyzed physiological data such as heart rate variability (HRV) and respiratory sinus arrhythmia (RSA) using MindWare Technologies software

TEACHING EXPERIENCE

Cornell University

2024, 2025, 2026

Graduate TA

- * Teaching Assistant for CS 4670 (Introduction to Computer Vision) – Spring 2026
- * Teaching Assistant for CS 2110 (Object-Oriented Programming and Data Structures) – Fall 2025
- * Teaching Assistant for CS 1110 (Introduction to Computing: A Design and Development Perspective) – Fall 2024

Howard University

Aug. 2021 – May 2022

Undergraduate TA

- * Teaching Assistant for CSCI 100 (Introduction to Computer Science)

SKILLS

Python, PyTorch, OpenCV, NumPy, Pandas, C++, C#, JavaScript, HTML/CSS, Java, L^AT_EX

VOLUNTEERING, FELLOWSHIPS, AWARDS, AND INTERESTS

Cornell SoNIC Program

June 2024

RA & TA

- * Program designed to help students learn about a variety of state-of-the-art computing research directions happening at Cornell University and beyond, and get an opportunity to conduct a short research project.
- * Introduced students to multimodal LLMs and gave guidance on the direction of their projects.

Cornell University CS Visit Day

Mar. 2024

Volunteer

CodeAfrique

Jan. 2024

Mentor

- * Program designed to help students in Ghana gain a window into the field of Computer Science
- * Taught students the fundamentals of Python and HTML, going over concepts such as functions, variables, loops and if-else statements

GEM Fellowship

Aug. 2022

Recipient

Cornell University Hopper Dean/Bowers CIS Dean's Excellence Fellowship

Aug. 2022

Recipient

International Conference on Applied Human Factors and Ergonomics (AHFE) 2020 Best Paper

Jul. 2020

Winner

- * Awarded Best Paper of 2020 for research paper *Evoking Stress Reactivity in a Virtual Dance Competition*
- * Written at Iowa State University during Summer 2019 internship

Karsh STEM Scholars Program*Member*

June 2018 – May 2022

- * Selected to be one of 29 students to be a part of the second cohort of Howard University's research PhD track development program
- * Participated in a summer bridge program on college success, research, and public speaking seminars
- * Engaged in activities that promoted group collaboration skills

Tau Beta Pi

May 2022 – Present

Member

Washington, DC

Upsilon Pi Epsilon

Mar. 2021 – Present

Member

Washington, DC

National Society of Black Engineers (NSBE)

Fall 2018 – Present

Member

Washington, DC

NSBE Jr.

Sep. 2018 – May 2019

Mentor

Washington, DC

- * Mentor for Tri-Math-a-Lon, VEX Robotics, and First Lego League
- * Dedicated to mentoring and tutoring young Black students in the greater D.C. area

Microsoft Code Academy

Dec. 2018 – May 2019

Mentor

Washington, DC

- * Taught a bimonthly class of secondary school students the principles of programming with the help of Microsoft employees
- * Provided mentoring and guidance to students outside the classroom

Association for Computing Machinery

Fall 2019 – Present

Member

Washington, DC